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1) Configuring a Basic Spring Application

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="
        http://maven.apache.org/POM/4.0.0
        http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.library</groupId>
    <artifactId>LibraryManagement</artifactId>
    <version>1.0</version>
    <dependencies>
        <!-- Spring Core -->
        <dependency>
            <groupId>org.springframework</groupId>
            <artifactId>spring-context</artifactId>
            <version>6.1.8</version>
        </dependency>
    </dependencies>
</project>
```

BookRepository.java

```
package com.library.repository;

public class BookRepository {

    public void getBookDetails() {

        System.out.println("Book details retrieved from the repository.");

    }
}
```

```
}
```

BookService.java

```
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {

    private BookRepository bookRepository;

    public void setBookRepository(BookRepository bookRepository) {

        this.bookRepository = bookRepository;

    }

    public void displayBook() {

        System.out.println("Book Service is Executing...");

        bookRepository.getBookDetails();

    }

}
```

applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"

    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

    xsi:schemaLocation="

        http://www.springframework.org/schema/beans

        https://www.springframework.org/schema/beans/spring-beans.xsd">

    <!-- Repository Bean -->

    <bean id="bookRepository"

        class="com.library.repository.BookRepository"/>

    <!-- Service Bean -->

    <bean id="bookService"

        class="com.library.service.BookService">

        <property name="bookRepository"
```

```
        ref="bookRepository"/>
    </bean>
</beans>
```

MainApplication.java

```
package com.library;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import com.library.service.BookService;

public class LibraryManagementApplication {

    public static void main(String[] args) {

        ApplicationContext context =

            new ClassPathXmlApplicationContext("applicationContext.xml");

        BookService service =

            (BookService) context.getBean("bookService");

        service.displayBook();

    }

}
```

2) Implementing Dependency Injection

BookRepository.java

```
package com.library.repository;

public class BookRepository {

    public void displayBookDetails() {

        System.out.println("Fetching book information from the repository...");

    }

}
```

BookService.java

```
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {

    private BookRepository bookRepository;

    public void setBookRepository(BookRepository bookRepository) {

        this.bookRepository = bookRepository;
    }

    public void showBookDetails() {

        System.out.println("Book Service is processing the request...");

        bookRepository.displayBookDetails();
    }
}
```

applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"

    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

    xsi:schemaLocation="

        http://www.springframework.org/schema/beans

        https://www.springframework.org/schema/beans/spring-beans.xsd">

    <!-- Book Repository Bean -->

    <bean id="bookRepository" class="com.library.repository.BookRepository"/>

    <!-- Book Service Bean -->

    <bean id="bookService" class="com.library.service.BookService">

        <property name="bookRepository"

            ref="bookRepository"/>

    </bean>

</beans>
```

LibraryManagementApplication.java

```
package com.library;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import com.library.service.BookService;

public class LibraryManagementApplication {

    public static void main(String[] args) {

        ApplicationContext context =

            new ClassPathXmlApplicationContext("applicationContext.xml");

        BookService service =

            (BookService) context.getBean("bookService");

        service.showBookDetails();

    }

}
```

3) Creating and Configuring a Maven Project

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

    xsi:schemaLocation="
        http://maven.apache.org/POM/4.0.0
        http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com. library </groupId>
    <artifactId>LibraryManagement</artifactId>
    <version>1.0-SNAPSHOT</version>

    <dependencies>

        <dependency>
            <groupId>org.springframework</groupId>
```

```
        <artifactId>spring-context</artifactId>
        <version>5.3.30</version>
    </dependency>

    <dependency>
        <groupId>org.springframework</groupId>
        <artifactId>spring-aop</artifactId>
        <version>5.3.30</version>
    </dependency>

    <dependency>
        <groupId>org.springframework</groupId>
        <artifactId>spring-webmvc</artifactId>
        <version>5.3.30</version>
    </dependency>
</dependencies>

<build>

    <plugins>

        <plugin>

            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-compiler-plugin</artifactId>
            <version>3.11.0</version>

            <configuration>

                <source>1.8</source>
                <target>1.8</target>

            </configuration>

        </plugin>

    </plugins>

</build>
```

</project>